

## No. of Functional Features Included in the Solution

|                      |                                 |
|----------------------|---------------------------------|
| <b>Date</b>          | 03 July 2026                    |
| <b>Team ID</b>       | SWTID-2026-8076                 |
| <b>Project Name</b>  | Credit Card Approval Prediction |
| <b>Maximum Marks</b> | 5 Marks                         |

### Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

|   | Feature Name                    | Feature Description                                            | Module / Component | Status | Marks Contribution |
|---|---------------------------------|----------------------------------------------------------------|--------------------|--------|--------------------|
| 1 | Applicant Details Form          | Collects applicant personal, employment and financial details. | User Interface     | Done   | High               |
| 2 | Input Validation                | Validates applicant information before prediction.             | Backend            | Done   | High               |
| 3 | Data Preprocessing              | Cleans and prepares applicant data for prediction.             | Machine Learning   | Done   | High               |
| 4 | Feature Encoding                | Converts categorical data into numerical values.               | Machine Learning   | Done   | High               |
| 5 | Feature Scaling                 | Normalizes numerical features using saved scaler.              | Machine Learning   | Done   | High               |
| 6 | Credit Card Approval Prediction | Predicts approval or rejection using trained ML model.         | Prediction Engine  | Done   | High               |
| 7 | Result Display                  | Displays prediction result on the web page.                    | User Interface     | Done   | Medium             |
| 8 | Model Loading                   | Loads trained model, encoder and scaler files.                 | Flask Backend      | Done   | Medium             |



| Metric                     | Count / Value |
|----------------------------|---------------|
| Total Features Planned     | 8             |
| Total Features Implemented | 8             |
| Core / Must-Have Features  | 8             |
| Additional Features        | 0             |
| Features Tested & Verified | 8             |

| s.no | Category          | Features in Category | Example Features                             |
|------|-------------------|----------------------|----------------------------------------------|
| 1    | User Interface    | 2                    | Applicant Details Form, Result Display       |
| 2    | Backend / Logic   | 4                    | Validation, Preprocessing, Encoding, Scaling |
| 3    | Machine Learning  | 1                    | Credit Card Approval Prediction              |
| 4    | Model Integration | 1                    | Flask + best_model.pkl                       |